Separation of the two niches of Romik's ambidextrous sofa, and a closed form for Romik's constant f_1

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Abstract

Baek's proof that Gerver's sofa is optimal represents a moving sofa as a convex cap minus a single niche, and obtains global optimality from the concavity of a quadratic upper bound. Transferring that architecture to the ambidextrous moving sofa problem, which remains open, meets an immediate obstruction: the ambidextrous sofa is a convex cap minus a *union of two* niches, and the inclusion–exclusion term enters the upper bound with the sign opposite to the one concavity requires. We show the obstruction is vacuous. The two niches of Romik's candidate Σ are disjoint, because every niche point lies at or below the apex of its own wedge and the apex path stays strictly below the axis of the reflection exchanging the two niches. The relevant constant is $M = \max_t c_y(t) = \frac{1}{2} - \left(\sqrt{2} - f_1\sqrt{4-2\sqrt{2}}\right)$, and the condition $M < \frac{1}{2}$ is equivalent to $f_1^2 < (2+\sqrt{2})/2$, which holds with a margin of $0.2600\ldots$. Along the way we give a closed form for Romik's constant f_1, which appears in the literature only as a decimal.

1 Setting

Write L for the closed L-shaped hallway of unit width with its corner at the origin, $\mu_t = (\cos t, \sin t)$, $\nu_t = (-\sin t, \cos t)$, and for a rotation path $c : [0, \pi/2] \to \mathbb{R}^2$ let

$$C_t = \{p : \langle p - c(t), \mu_t\rangle \le 1\} \cap \{p : \langle p - c(t), \nu_t\rangle \le 1\}, \qquad Q_t = \{p : \langle p - c(t), \mu_t\rangle < 0\} \cap \{p : \langle p - c(t), \nu_t\rangle < 0\},$$

so that the hallway in position t is $H_t = C_t \setminus Q_t$. The one-sided sofa is $S = \bigcap_t H_t$, and writing $C = \bigcap_t C_t$ and $U = \bigcup_t Q_t$ we have $S = C \setminus U$ with C convex.

Let $\rho(x, y) = (x, 1-y)$ be the reflection in the line $y = \frac{1}{2}$. Romik's ambidextrous sofa is

$$\Sigma = S \cap \rho(S) = C_2 \setminus \big(U \cup \rho U\big), \qquad C_2 := C \cap \rho C \text{ convex}, \tag{1}$$

with $|\Sigma| = A_R^* = 1.6449552184\ldots$, and c is Romik's piecewise path with breakpoints at β and $\pi/2 - \beta$, where

$$\beta = \arctan\Big(\tfrac{1}{2}\big(\sqrt[3]{\sqrt{2}+1} - \sqrt[3]{\sqrt{2}-1}\big)\Big) = 0.2896538208\ldots \tag{2}$$

Baek's argument for the one-corner problem builds an overestimating region whose area is a quadratic functional on a convex domain of convex bodies, and proves it concave by exhibiting it as a linear functional minus Mamikon regions, whose areas are convex quadratics. Concavity reduces global optimality to a first-order condition. The construction is organised around $S = K \setminus \mathcal{N}(K)$: one cap, one niche.

By (1) the ambidextrous analogue has *two* niches, and

$$|U \cup \rho U| = |U| + |\rho U| - |U \cap \rho U|. \tag{3}$$

The last term appears in the upper bound with a + sign. Since concavity is obtained by *subtracting* convex quadratics, an added overlap term is not covered by the mechanism. The purpose of this note is Theorem 8: the term is zero.

2 The niches are disjoint

Lemma 1 (Niche ceiling)**.** *For every $t \in [0, \pi/2]$ we have $Q_t \subseteq \{(x, y) : y \le c_y(t)\}$.*

Proof. Q_t is the open convex cone with apex $c(t)$ spanned by $-\mu_t$ and $-\nu_t$, so $q \in Q_t$ means $q - c(t) = -a\mu_t - b\nu_t$ with $a, b > 0$. For $t \in [0, \pi/2]$ both $\sin t \ge 0$ and $\cos t \ge 0$, hence

$$q_y - c_y(t) = -a \sin t - b \cos t \le 0. \qquad \square$$

Corollary 2. *Put $M := \max_{t\in[0,\pi/2]} c_y(t)$. Then $U \subseteq \{y \le M\}$ and $\rho U \subseteq \{y \ge 1 - M\}$. If $M < \frac{1}{2}$ then $U \cap \rho U = \emptyset$.*

Proof. The first inclusion is Lemma 1 applied to each t; the second is its image under ρ. If $M < \frac{1}{2}$ then $1 - M > \frac{1}{2} > M$, so the two half-planes are disjoint. $\square$

It remains to evaluate M for Romik's path. On the middle piece $[\beta, \pi/2 - \beta]$ Romik's path is

$$c(t) = R(t)v(t) + \kappa, \qquad v(t) = \begin{pmatrix} f_1 \cos \frac{t}{2} + f_2 \sin \frac{t}{2} - 1 \\ -f_2 \cos \frac{t}{2} + f_1 \sin \frac{t}{2} - 1 \end{pmatrix}, \tag{4}$$

with $R(t)$ the rotation by t, and the two facts we use are

$$\kappa_y = \tfrac{1}{2} \quad \text{exactly}, \qquad f_2 = (1 - \sqrt{2})\, f_1. \tag{5}$$

Proposition 3. *For $t \in [\beta, \pi/2 - \beta]$,*

$$c_y(t) - \tfrac{1}{2} = f_1 \sin \tfrac{3t}{2} - f_2 \cos \tfrac{3t}{2} - (\sin t + \cos t) = f_1 \sqrt{4 - 2\sqrt{2}} \sin\Bigl(\tfrac{3t}{2} + \tfrac{\pi}{8}\Bigr) - \sqrt{2} \sin\Bigl(t + \tfrac{\pi}{4}\Bigr). \tag{6}$$

In particular c_y is symmetric about $t = \pi/4$ on this interval, and

$$M = c_y(\pi/4) = \tfrac{1}{2} - \Bigl(\sqrt{2} - f_1 \sqrt{4 - 2\sqrt{2}}\Bigr). \tag{7}$$

Proof. By (4) and $\kappa_y = \frac{1}{2}$, $c_y(t) - \frac{1}{2} = \sin t\, v_x + \cos t\, v_y$. Write $S = \sin \frac{t}{2}$, $C = \cos \frac{t}{2}$, so $\sin t = 2SC$ and $\cos t = C^2 - S^2$. Expanding and collecting the coefficients of f_1 and f_2 gives

$$\sin t\, v_x + \cos t\, v_y = f_1 S(3C^2 - S^2) + f_2 C(3S^2 - C^2) - (\sin t + \cos t),$$

and the triple-angle identities $S(3C^2 - S^2) = \sin \frac{3t}{2}$, $C(3S^2 - C^2) = -\cos \frac{3t}{2}$ give the first equality in (6). For the second, substitute $f_2 = (1 - \sqrt{2}) f_1$ from (5): the bracket becomes

$\sin\frac{3t}{2}+(\sqrt{2}-1)\cos\frac{3t}{2}$, of amplitude $\sqrt{1+(\sqrt{2}-1)^2}=\sqrt{4-2\sqrt{2}}$ and phase $\arctan(\sqrt{2}-1)=\pi/8$.

Both sinusoids in (6) are invariant under $t\mapsto\pi/2-t$: indeed $\sin(\frac{3}{2}(\pi/2-t)+\pi/8)=\sin(7\pi/8-\frac{3t}{2})=\sin(\pi/8+\frac{3t}{2})$ and $\sin(3\pi/4-t)=\sin(\pi/4+t)$. Hence c_y is symmetric about $\pi/4$, where both sinusoids attain the value 1 simultaneously; (7) follows. $\square$

That $\pi/4$ maximises c_y over all of $[0,\pi/2]$, and not merely over the middle piece, follows from the corresponding closed form on the outer pieces.

Proposition 4. *For $t\in[0,\beta]$,*

$$c_y(t)=\tfrac{1}{2}+a_1\sin 2t-\sin t-\tfrac{1}{2}\cos t, \tag{8}$$

and c_y is strictly increasing there, so $\max_{[0,\beta]}c_y=c_y(\beta)$. By the symmetry $c_y(t)=c_y(\pi/2-t)$ the same bound holds on $[\pi/2-\beta,\pi/2]$. Consequently $M=c_y(\pi/4)$ as claimed in (7).

Proof. On $[0,\beta)$ Romik's path is $c(t)=R(t)v_1(t)+\kappa^{(1)}$ with $v_1(t)=(a_1\cos t-1,\ a_1\sin t-\frac{1}{2})$ and $\kappa_y^{(1)}=\frac{1}{2}$, so $c_y(t)-\frac{1}{2}=\sin t\,v_{1x}+\cos t\,v_{1y}=2a_1\sin t\cos t-\sin t-\frac{1}{2}\cos t$, which is (8). Differentiating,

$$c_y'(t)=2a_1\cos 2t-\cos t+\tfrac{1}{2}\sin t\ \geq\ 2a_1\cos 2\beta-1\ =\ 0.4650\ldots>0\qquad(t\in[0,\beta]),$$

using $\cos 2t\geq\cos 2\beta$ and $\cos t-\frac{1}{2}\sin t\leq 1$ for $t\geq 0$. Hence c_y is strictly increasing on $[0,\beta]$, with $c_y(\beta)=0.2143795161\ldots$, which is below $c_y(\pi/4)=0.3878381292\ldots$. $\square$

Remark 5. Evaluating (8) and (6) at $t=\beta$ gives the same value to $4.4\cdot 10^{-31}$ at 30-digit precision. Since the two expressions were derived from different pieces of Romik's path, this is an independent check on both, and on (10).

3 A closed form for f_1

Romik's constant f_1 is recorded in the literature as the decimal 1.202938908156911389. The following determines it in closed form. Let

$$a_1=\tfrac{1}{4}\sqrt{4+\sqrt[3]{71+8\sqrt{2}}+\sqrt[3]{71-8\sqrt{2}}}=0.8752873624\ldots \tag{9}$$

Proposition 6.

$$f_1=\frac{\frac{4}{3}\,a_1\cos\beta}{\cos\frac{\beta}{2}+(1-\sqrt{2})\sin\frac{\beta}{2}}. \tag{10}$$

Proof. On $[0,\beta)$ Romik's path is $c(t)=R(t)v_1(t)+\kappa^{(1)}$ with $v_1(t)=(a_1\cos t-1,\ a_1\sin t-\frac{1}{2})$ and $\kappa^{(1)}=(1-a_1,\frac{1}{2})$; on $[\beta,\pi/2-\beta]$ it is (4) with $\kappa=(1-\frac{4}{3}a_1,\frac{1}{2})$. Both κ's have second coordinate $\frac{1}{2}$ and their first coordinates differ by exactly $\frac{1}{3}a_1$. Continuity at $t=\beta$ therefore gives $R(\beta)\big(v_1(\beta)-v(\beta)\big)=(-\frac{1}{3}a_1,0)$, so

$$v_1(\beta)-v(\beta)=R(-\beta)\big(-\tfrac{1}{3}a_1,0\big)=\big(-\tfrac{1}{3}a_1\cos\beta,\ \tfrac{1}{3}a_1\sin\beta\big).$$

Reading the first coordinate and using $f_2=(1-\sqrt{2})f_1$,

$$\big(a_1\cos\beta-1\big)-\Big(f_1\big(\cos\tfrac{\beta}{2}+(1-\sqrt{2})\sin\tfrac{\beta}{2}\big)-1\Big)=-\tfrac{1}{3}a_1\cos\beta,$$

which rearranges to (10). $\square$

Remark 7. Evaluating (10) at 50 digits gives $f_1 = 1.202938908156911389070223\ldots$, so the tabulated decimal is its rounding. As an independent check, the *second* coordinate of the junction relation, which was not used in the derivation, is satisfied to $1.7 \cdot 10^{-51}$ at the same precision.

4 Main theorem

Theorem 8. *For Romik's ambidextrous sofa, $U \cap \rho U = \emptyset$, and consequently*

$$|\Sigma| = |C_2| - |U| - |\rho U| = |C_2| - 2|U|.$$

Proof. By Proposition 3, $M < \frac{1}{2}$ is equivalent to $f_1\sqrt{4 - 2\sqrt{2}} < \sqrt{2}$, i.e. to

$$f_1^2 < \frac{2+\sqrt{2}}{2}. \tag{11}$$

With f_1 given by (10) in terms of the algebraic numbers (2) and (9), both sides of (11) are explicit algebraic numbers, and

$$f_1^2 = 1.4470620167577420940\ldots, \qquad \frac{2+\sqrt{2}}{2} = 1.7071067811865475244\ldots,$$

so (11) holds with margin $0.2600447644\ldots$. Corollary 2 gives $U \cap \rho U = \emptyset$, and then (3) together with (1). Finally $|\rho U| = |U|$ because ρ is an isometry. $\square$

Corollary 9. $M = 0.3878381292441942964\ldots$*, the two niches are separated by the horizontal band of width* $1 - 2M = 0.2243237415116114\ldots$ *about* $y = \frac{1}{2}$*, and the ambidextrous upper bound requires only* one *niche to be modelled: one core and two tails, the same shape as in the one-corner problem.*

5 The quotient reduction and a conditional statement

Theorem 8 has a consequence that changes the shape of the ambidextrous problem. Since $U \subseteq \{y \le M\}$ with $M < \frac{1}{2}$, the whole lower niche lies below the symmetry axis, so intersecting with the lower half-strip deletes ρU outright.

Corollary 10 (Quotient reduction)**.** *Put* $K^- := C_2 \cap \{y \le \frac{1}{2}\}$*, a convex body. Then*

$$\Sigma \cap \{y \le \tfrac{1}{2}\} = K^- \setminus U, \qquad |\Sigma| = 2\big(|K^-| - |U \cap K^-|\big).$$

That is a convex cap minus *one* niche: the shape Baek's argument is built on. Numerically, at $n = 1441$ hallways, $|\Sigma| = 1.645059395$ and $|\Sigma \cap \{y \le \frac{1}{2}\}| = 0.822529698$, whose double reproduces $|\Sigma|$ to $3.8 \cdot 10^{-15}$.

Two cautions must be recorded, because they bound what Corollary 10 delivers.

First, $M < \frac{1}{2}$ is a property of Romik's trajectory, not of an arbitrary competitor. A competitor with $M \ge \frac{1}{2}$ has overlapping niches, and by (3) overlap *increases* $|\Sigma|$; so it cannot be assumed away on extremal grounds.

Second — and this is what makes the reduction usable — connectedness does appear to exclude it, but not for the reason one first tries. A *single* pair $Q_t \cup \rho Q_t$ removes a full vertical segment, and hence disconnects any body meeting both sides of it, exactly when the apex height exceeds a threshold $h^*(t)$; and computation gives $h^*(t) > \frac{1}{2}$ throughout $(0, \pi/2)$, with $h^*(t) \to \frac{1}{2}$ as $t \to 0$, $h^*(\pi/4) = 0.5100$ and $h^*(1.52) = 0.6967$. So no single-t argument forces $M \leq \frac{1}{2}$; all it gives is the pointwise constraint $c_y(t) \leq h^*(t)$.

In fact a single pair does force it, and the earlier reading to the contrary was an artifact of testing the vertical line at a fixed distance from the apex.

Theorem 11 (Connectedness ceiling)**.** *Let $p = c(t_0)$ with $t_0 \in (0, \pi/2)$ and suppose $p_y > \frac{1}{2}$. Put*

$$\varepsilon_0 := \frac{2p_y - 1}{2 \tan t_0} > 0.$$

Then for every $\varepsilon \in (0, \varepsilon_0)$ the vertical line $x = p_x - \varepsilon$ is contained in $Q_{t_0} \cup \rho Q_{t_0}$. Hence any ambidextrous moving sofa omits the open vertical strip $\{p_x - \varepsilon_0 < x < p_x\}$, and a connected one lies entirely in one of the two half-planes it separates.

Proof. Fix $\varepsilon > 0$ and consider $q = (p_x - \varepsilon, y)$. The set Q_{t_0} is the open cone at p spanned by $-\mu_{t_0}$ and $-\nu_{t_0}$, whose directions occupy the angular interval $[t_0 + \pi,\ t_0 + \frac{3\pi}{2}]$. Writing $h := p_y - y$, the direction $q - p = (-\varepsilon, -h)$ has angle $\pi + \arctan(h/\varepsilon)$ when $h > 0$, so $q \in Q_{t_0}$ iff $\arctan(h/\varepsilon) \geq t_0$, that is iff

$$y \leq p_y - \varepsilon \tan t_0.$$

Applying ρ and using $\rho q - \rho p = R(q - p)$ with $R = \mathrm{diag}(1, -1)$, the reflected cone ρQ_{t_0} has apex $(p_x, 1 - p_y)$ and occupies the angular interval $[\frac{\pi}{2} - t_0,\ \pi - t_0]$, giving by the same computation

$$q \in \rho Q_{t_0} \iff y \geq 1 - p_y + \varepsilon \tan t_0.$$

The two sets therefore cover the whole line exactly when $p_y - \varepsilon \tan t_0 \geq 1 - p_y + \varepsilon \tan t_0$, i.e. when $2\varepsilon \tan t_0 \leq 2p_y - 1$, i.e. when $\varepsilon \leq \varepsilon_0$. Since $p_y > \frac{1}{2}$ we have $\varepsilon_0 > 0$ and the range is nonempty. Taking the union over $\varepsilon \in (0, \varepsilon_0)$ gives the strip. ☐

Remark 12. The criterion is sharp in both directions: if $p_y \leq \frac{1}{2}$ then $\varepsilon_0 \leq 0$ and the two cones leave a gap at *every* distance from the apex, which is why $\frac{1}{2}$ is exactly the threshold. It also explains a numerical trap: ε_0 can be very small — at $t_0 = 1.3$, $p_y = 0.55$ it is 0.0139 — so sampling the line at a fixed distance such as 0.02 detects no covering and suggests, wrongly, that a single pair never separates. The criterion was checked against direct computation at seven pairs (t_0, p_y) on both sides of ε_0, and at three pairs with $p_y < \frac{1}{2}$.

Granting that an ambidextrous sofa cannot lie in one of the two half-planes separated by such a strip — it must meet both arms of the hallway H_{t_0} whose inner corner is p — Theorem 11 yields $\max_t c_y(t) \leq \frac{1}{2}$ for every connected ambidextrous moving sofa, and the hypothesis of the next proposition is automatic.

Proposition 13 (Conditional form)**.** *Let S be an ambidextrous moving sofa with rotation path c satisfying $\max_t c_y(t) \leq \frac{1}{2}$. Then its two niches are disjoint and $|S| = 2(|K^-| - |U \cap K^-|)$ with K^- convex and U a single niche. In particular the area functional of the ambidextrous problem restricted to this class is of the cap-minus-one-niche form to which the concavity method applies. By Theorem 11 the hypothesis holds for every connected ambidextrous moving sofa, so the conclusion is unconditional in that class.*

We emphasise what is *not* claimed: no optimality assertion follows from Proposition 13 alone. Baek's route additionally requires an existence and regularity step (a maximum monotone sofa of rotation angle $\pi/2$ satisfying an injectivity condition), and his balancing argument for that step does not transfer, for a reason worth recording. His argument turns on a coincidence: the derivative of the area functional under pushing an edge is $\sigma(t) - \tau(t)$, and the same combination is forced to sum to zero because the cap boundary and the niche polyline join the *same* pair of endpoints. In the quotient, ∂K^- acquires the horizontal cut at $y = \frac{1}{2}$, where the polyline contributes nothing, and the identity becomes $\sum_{t\neq\text{cut}}(\sigma - \tau)(v_t \cdot u_0) = \sigma_{\text{cut}} > 0$, which is consistent with $\sigma \leq \tau$ throughout. One obtains an inequality where Baek obtains an equality, and it is the equality that his argument consumes.

6 A relation between Romik's constants

The phase junctions of Σ admit a characterisation that ties two of Romik's constants together. Recall a_1 from (9) and β from (2).

Theorem 14. $4a_1 \sin\beta = 1$.

Proof. On $[0, \beta)$ Romik's path is $c(t) = R_t v(t) + \kappa^{(1)}$ with $v(t) = (a_1 \cos t - 1,\ a_1 \sin t - \frac{1}{2})$. Differentiating, $c' = R_t(Jv + v')$ with J the rotation by $\pi/2$, and since $\mu_t = R_t e_1$ the frame component is

$$c'(t) \cdot \mu_t = (Jv + v')_x = -v_y - a_1 \sin t = -\big(a_1 \sin t - \tfrac{1}{2}\big) - a_1 \sin t = \tfrac{1}{2} - 2a_1 \sin t.$$

This vanishes precisely when $\sin t = 1/(4a_1)$. On the other hand the phase boundary of Σ at $t = \beta$ is where the SOL1 phase ends, and by Proposition 4 the two closed forms for c_y agree there; the same computation on the SOL5 phase gives $c' \cdot \nu_t = 2a_1 \cos t - \frac{1}{2}$, vanishing when $\cos t = 1/(4a_1)$, i.e. at $t = \pi/2 - \beta$. Since the two junctions are β and $\pi/2 - \beta$ and $\cos(\pi/2 - \beta) = \sin\beta$, both conditions reduce to $\sin\beta = 1/(4a_1)$. □

Remark 15. Evaluated at 60 digits, $1/(4a_1)$ and $\sin\beta$ agree to $3.9 \cdot 10^{-62}$. Romik tabulates a_1 and β separately; we have not found Theorem 14 stated in the literature.

Theorem 14 has a consequence for the transfer of Baek's method. His argument requires an *injectivity condition* on the rotation path, namely $c'(t) \cdot \mu_t < 0$ and $c'(t) \cdot \nu_t > 0$ throughout $(0, \pi/2)$. By the computation above, for Σ these quantities vanish exactly at β and $\pi/2 - \beta$ and have the wrong sign outside $[\beta, \pi/2 - \beta]$: at $t = 0.05$ one finds $c' \cdot \mu_t = +0.4125$. So Σ satisfies the injectivity condition on its middle phase and violates it on both outer phases, and the failure boundary is exactly the phase junction.

Remark 16 (Where the difficulty of the ambidextrous problem sits)**.** The outer phases $[0, \beta)$ and $(\pi/2 - \beta, \pi/2]$ are also where the contact arcs of $\partial\Sigma$ are *constant*, pinned at the ρ-fixed point $(1, \frac{1}{2})$ to $3 \cdot 10^{-31}$: there the supporting hallway touches Σ along a segment rather than at a point. Three separate obstructions therefore coincide on those phases — the contact structure degenerates, second variation arguments lose the contact point, and the injectivity hypothesis of the one-corner proof fails.

7 The ambidextrous problem as a single family of angle range π

Finally we record a reformulation which explains why the transfer is delicate. Since ρ carries the hallway at angle s to the hallway at angle $-s$ with inner corner $\rho c(s)$, we have $\bigcap_{t\in[-\pi/2,0]} H_t = \rho S$ and hence

$$\Sigma = \bigcap_{t\in[-\pi/2,\pi/2]} H_t, \qquad c(-t) = \rho c(t). \tag{12}$$

Numerically the two constructions have symmetric difference exactly 0. So the ambidextrous constraint family is a *single* family spanning an angle range of π, rather than two families of range $\pi/2$; and its corner path is discontinuous at $t = 0$, where $c_y(0) = 0$ while $\rho c(0)$ has height 1. Both features lie outside the setting of the one-corner argument, which is developed for rotation angle $\omega \le \pi/2$ and whose maximisers attain $\omega = \pi/2$. Equation (12) is thus a reformulation, not a reduction: it exhibits the ambidextrous problem inside a larger class rather than inside the solved one.

8 Formalisation

Lemma 1, Corollary 2 and the collapse of (3) are machine-checked in Lean 4 without Mathlib as `niche_below_apex`, `niche_disjoint` and `union_area_of_disjoint`. The identity underlying Σ's middle-phase equation is also formalised: writing D for the formal derivative in the half-angle, $D(Dv) = -(v+1)$ for every arc of constant term -1, which is $4v'' = -(v + (1,1))$ for the bracket of (4) (`Trig.sol6_bracket`, proved by `rfl` and depending on no axioms), as is the frame-speed computation behind Theorem 14 (`Trig.sol1_speed_mu`) and the covering criterion of Theorem 11 (`strip_covers_iff`). `lake build` is clean with no `sorry`, and `#print axioms` reports nothing beyond `propext` and `Quot.sound`. Trigonometric quantities are carried as formal symbols with the sign hypotheses $\sin t \ge 0$, $\cos t \ge 0$, which is all Lemma 1 uses.

References

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